

Man-in-the-Middle and Key Recovery Attacks against QP-KEM

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Abstract. The Q-problem has been introduced as a new post-quantum hard problem for the development of new post-quantum cryptography. We present two man-in-the-middle and four key recovery attacks against the key exchange protocol based on the Q-problem. The man-in-the-middle attacks take negligible time and allow the attacker to recover the exchanged key. The most effective key recovery attack is computationally trivial and, unlike the other attacks presented in this work, cannot be prevented by countermeasures.

Keywords: Q-problem · QP-KEM · key exchange protocol · PQC · post-quantum cryptography

1 Introduction

Advances in the development of quantum computers threaten the security of widely used cryptography. To protect information, new post-quantum key exchange methods have been proposed [AAA⁺19]. One of these key exchange methods is a protocol based on the Q-problem: QP-KEM [KKP⁺25]. We present two efficient Man-in-the-Middle (MitM) attacks and four key recovery attacks against QP-KEM. To the best of our knowledge, this paper presents the first attacks against QP-KEM.

The Q-problem was introduced by Laouid, Kara, Hammoudeh, and Al-Essa, which they defined as [LKHA24]:

Q-problem. Given p and z , find x and y where $x \star y \bmod p = z$.

In the Q-problem, $\star$ denotes an operator, for example $+$, $\times$ or exp . The challenge in the Q-problem is to find the correct x and y from many possible combinations of x and y resulting in the same z . However, the problem does not define a correct hardness assumption as it is impossible to find the correct solution due to lack of information. A key exchange cannot be directly based on the Q-problem as a recipient in a protocol given p and z cannot distinguish the correct (x, y) out of many possibilities. To work around this limitation of the Q-problem, the key exchange based on the Q-problem, QP-KEM, is a multi-round interactive key exchange protocol where the security is based on the complexity of the protocol rather than a hardness assumption of the Q-problem.

QP-KEM was presented by Kara, Karampidis, Panagiotakis, Hammoudeh, Felemban, and Papadourakis as a lightweight and efficient post-quantum Key Encapsulation Mechanism (KEM) [KKP⁺25]. This paper assumes the parameter sizes proposed in [KKP⁺25, Sec. 6], claimed to achieve 128-bit security. In addition to exchanging four 300 bit numbers,

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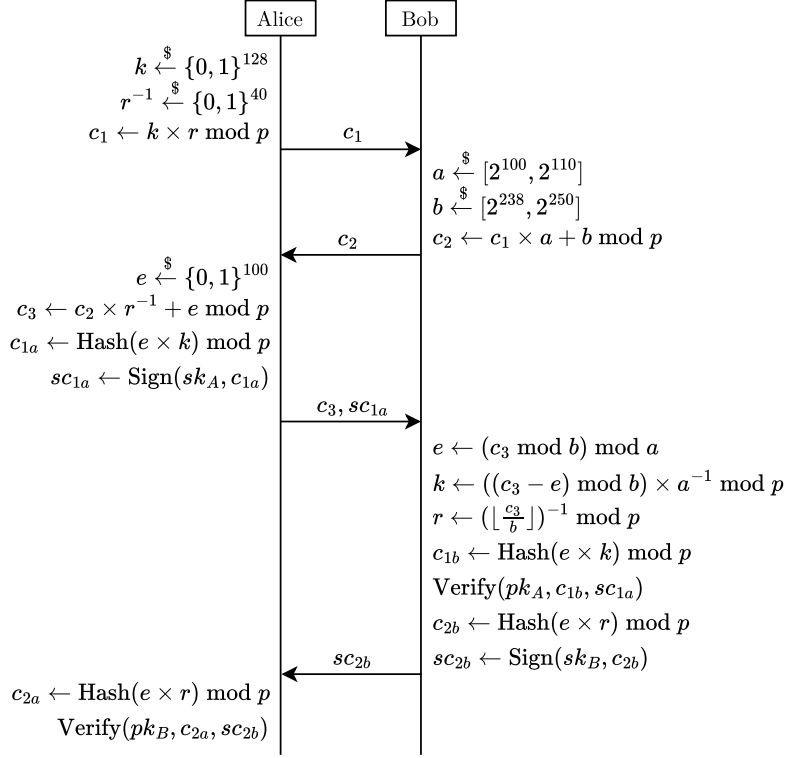


Figure 1: Key exchange using QP-KEM.

QP-KEM also exchanges two signatures to achieve mutual authentication. Assuming Falcon at a security level of 128-bit is used for these two signatures, QP-KEM requires to exchange 10 956 bits of information. QP-KEM also requires two round-trips, making the protocol incompatible with the definition of a KEM, and therefore we will refer to it as a key exchange protocol [CS01]. The complete key exchange protocol is shown in Figure 1, where r^{-1} indicates the modular multiplicative inverse of r . The protocol assumes that Alice and Bob have already exchanged public keys for authentication, pk_a and pk_b , and have agreed on a fixed 300 bit public safe prime p . The exchanged values c_1 , c_2 and c_3 are all calculated from an equation in which two variables should be unknown to the attacker, causing the attacker to find multiple potential solutions when trying to solve one of the equations. This prevents quantum period-finding algorithms from finding the correct solution.

Section 2, 3 and 4 describe the different attacks against QP-KEM. Section 2 shows the MitM attacks, section 3 shows the key recovery attacks using brute force, and section 4 shows a key recovery attack using lattice reduction. Finally, section 5 concludes the paper.

2 Man-in-the-middle attacks

In this section, we first present two MitM attacks, both of which require the attacker to be able to intercept and change messages between Alice and Bob. After presenting the attacks, we propose a countermeasure that prevents both attacks.

2.1 Malicious c_2

We show the first MitM attack in Figure 2. The attacker (Mallory) replaces c_2 from Bob by a value such that e can be recovered from c_3 independent of the value of r^{-1} . Any value between $2^{|e|} = 2^{100}$ and $2^{|p|-|r^{-1}|} = 2^{260}$ would suffice. Choosing 2^{100} causes Alice to compute:

$$mc_3 \leftarrow 2^{100} \times r^{-1} + e \mod p$$

However, since $e < 2^{100}$ and $2^{100} \times r^{-1} < 2^{140}$, it follows that $2^{100} \times r^{-1} + e < p$, the modular reduction thus has no effect and we have:

$$mc_3 = 2^{100} \times r^{-1} + e$$

This makes it possible to decompose r^{-1} and e from the value mc_3 that Alice sends to Bob:

$$\begin{aligned} e &\leftarrow mc_3 \mod 2^{100} \\ r^{-1} &\leftarrow \lfloor \frac{mc_3}{2^{100}} \rfloor \\ c_3 &\leftarrow c_2 \times r^{-1} + e \mod p \\ k &\leftarrow c_1 \times r^{-1} \mod p \end{aligned}$$

After recovering e and r^{-1} , Mallory sends c_3 to Bob. Alice and Bob then continue the protocol without having noticed Mallory who has recovered the key k .

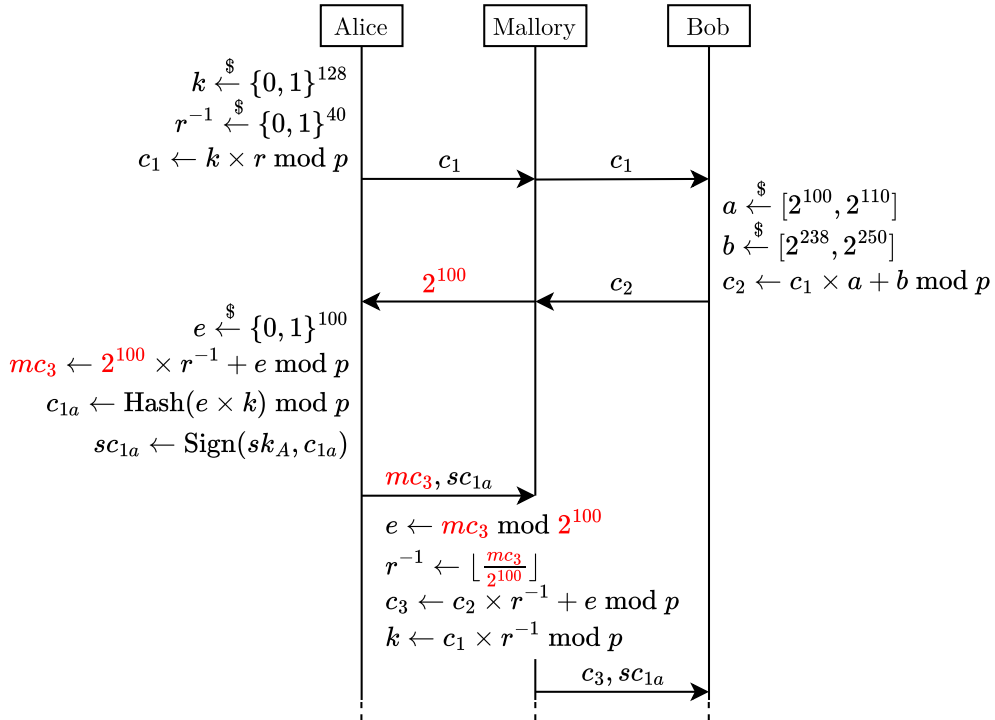


Figure 2: Man-in-the middle attack against QP-KEM. 2^{100} is chosen by Mallory and mc_3 is influenced by this value.

2.2 Bob signing for Mallory

The second attack is shown in Figure 3. Mallory forwards c_1 and after she has recovered e , r^{-1} and k , which were generated by Alice, she uses those values in a key exchange with Bob. Mallory tricks Bob into signing $e \times r \bmod p$ and forwards the signature to Alice. Alice and Bob receive a valid signature from each other and will continue the key exchange even though Mallory has recovered the key.

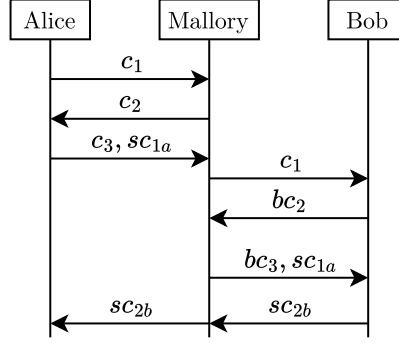


Figure 3: Man-in-the middle attack against QP-KEM. Mallory tricks Bob into signing $e \times r \bmod p$.

2.3 Countermeasure

We propose a countermeasure against both attacks presented in Section 2: sign and verify c_1 and c_2 . The concrete modification that would result in the least changes to the protocol would be to change:

$$c_{1a} \leftarrow \text{Hash}(e \times k) \bmod p$$

to:

$$c_{1a} \leftarrow \text{Hash}(e \times k \parallel c_1 \parallel c_2 \parallel \text{Bob}) \bmod p$$

and to change:

$$c_{2b} \leftarrow \text{Hash}(e \times r) \bmod p$$

to:

$$c_{2b} \leftarrow \text{Hash}(e \times r \parallel \text{Alice}) \bmod p$$

and to change the calculation of c_{1b} and c_{2a} accordingly.

This modification will allow Bob to detect any changes made to c_1 and c_2 , he can then notify Alice or terminate the protocol, preventing both attacks. Adding the identity of the intended recipient is necessary to prevent Mallory from using c_1 and c_2 exchanged between Alice and Mallory or between Bob and Mallory against Alice and Bob. Although signing c_1 is not necessary to prevent the attacks shown in Figure 2 and Figure 3, this prevents the attacker from setting c_1 to $2^{|b|}$, therefore learning b and 50 out of 110 bits of a . This would allow the attacker to guess the remaining 60 bits of a , reducing the security of QP-KEM from 128-bit to 60-bit.

Another countermeasure would be to send a signature of c_1 and c_2 together with the value. However, this would decrease the performance of the protocol as the number of signatures that must be generated and verified increases.

3 Brute-force key recovery attacks

QP-KEM aims to achieve a security level of 128-bit. However, we show that the key can be recovered in less than 2^{128} guesses. As soon as Alice and Bob start to use the key to authenticate messages, the adversary can check if a guess for the key is correct and can try to guess values for r^{-1} , e , and a . We first describe the attacks below and then propose countermeasures.

3.1 Key recovery in 2^{40} guesses for r^{-1}

Assuming an attacker intercepts the first message of the key exchange, c_1 , and intercepts a message m authenticated using a Message Authentication Code (MAC) with tag t , an attacker can recover the key by trying all possible values for r^{-1} , $0 \leq r^{-1} < 2^{40}$, using the following algorithm:

```

for  $r^{-1} := 0, \dots, 2^{40}$  do
   $k := c_1 \times r^{-1} \bmod p$ 
  if  $\text{MAC}_k(m) == t$  then
    return  $k$ 

```

Although there are multiple solutions for k and r^{-1} in $c_1 \leftarrow k \times r \bmod p$, as soon as the key k is used to authenticate, the attacker has a way to verify if a guessed key is correct.

3.2 Key recovery in 2^{100} guesses for e

Assuming that the attacker intercepts c_1 , c_2 , c_3 and an authenticated message, the key can also be recovered by trying all possible 2^{100} values for e .

```

for  $e := 0, \dots, 2^{100}$  do
   $r^{-1} := (c_3 - e) \times c_2^{-1} \bmod p$ 
   $k := c_1 \times r^{-1} \bmod p$ 
  if  $\text{MAC}_k(m) == t$  then
    return  $k$ 

```

3.3 Key recovery in 2^{110} guesses for a

Similarly to the previous two attacks, the key can be recovered by trying all 2^{110} possible values for a :

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for  $a := 0, \dots, 2^{110}$  do
   $b := c_2 - a \times c_1 \bmod p$ 
   $e := (c_3 \bmod b) \bmod a$ 
   $k := ((c_3 - e) \bmod b) \times a^{-1} \bmod p$ 
  if  $\text{MAC}_k(m) == t$  then
    return  $k$ 

```

3.4 Countermeasures

A countermeasure against the attack against r^{-1} (Section 3.1) is to increase the size of r^{-1} to 128 bits. p will then have to be increased to at least $2^{|b|+|r^{-1}|} = 2^{378}$ as p has to be

greater than $b \times r^{-1}$ (setting 4 in [KKP⁺25]) for Bob to be able to extract the key from c_3 . However, p will have to be increased even more to prevent the attack against e .

A countermeasure against the attack against e (Section 3.2) is to increase the size of e to at least 128 bits. However, since a has to be greater than e (setting 2), the minimum value of a has to be increased to at least the maximum value of e , 2^{128} , which also prevents the attack against a (Section 3.3). Since b has to be greater than $a \times k$ (setting 3), the minimum value of b has to be increased to at least 2^{257} . Setting the maximum value of b to 2^{258} requires to increase p to at least $2^{|b|+|r^{-1}|} = 2^{386}$. The new parameters are summarised in Table 1.

Table 1: Original and proposed values for QP-KEM in (except 0) $\log_2$. Maximum values are exclusive.

Param.	Orig. min.	Orig. max.	Pro. min.	Pro. max.
p	299	300	386	387
r⁻¹	0	40	0	128
e	0	100	0	128
a	100	110	128	129
b	238	250	257	258

4 Key recovery using lattice reduction

We first show the attack and then show that the attack cannot be prevented by increasing the size of the parameters in the protocol.

4.1 Attack

r^{-1} and e can be recovered from c_3 using c_2 and lattice reduction, which can then be used to recover the key k . This is possible as both r^{-1} and e are relatively small compared to p . A similar attack can be used to recover r^{-1} and k from c_1 when the greatest common divisor of r^{-1} and k is small. Recall the calculation of c_3 :

$$\begin{aligned} c_3 &= c_2 \times r^{-1} + e \pmod{p} \\ c_3 - e &= c_2 \times r^{-1} + p * y \end{aligned}$$

Consider the lattice spanned by the row vectors in the matrix:

$$\mathbf{B} = \begin{bmatrix} 1 & c_2 \\ 0 & p \end{bmatrix}$$

All vectors of coordinates in this lattice are of the form $(x, c_2 \times x + p \times y)$ for some integers x and y . For $x = r^{-1}$, the vector $(r^{-1}, c_3 - e)$ is a point in this lattice. Therefore, define the target vector $\mathbf{t}$ as:

$$\mathbf{t} = (0, c_3)$$

Then find the vector that is closest to $\mathbf{t}$ by solving the Closest Vector Problem (CVP). This can be efficiently solved using Babai's algorithm [Bab86].

$$(r^{-1}, c_3 - e) \leftarrow \text{ClosestVector}(\mathbf{t}, \mathbf{B})$$

r^{-1} and c_1 can then be used to recover the key k :

$$k \leftarrow c_1 \times r^{-1} \pmod{p}$$

4.2 Countermeasure

The attack cannot be prevented by increasing the size of parameters, as then the key size has to be decreased to an insecure small value. To prevent this specific attack, the sizes of e and r^{-1} must be increased so that the vector $\mathbf{u} = (r^{-1}, e)$ is not unusually short. The Gaussian heuristic tells us then that:

$$\begin{aligned} \|\mathbf{u}\| &> \sqrt{\det(\mathbf{B})} \\ \|\mathbf{u}\| &> \sqrt{p} \\ (r^{-1})^2 + e^2 &> p \end{aligned}$$

Assuming r^{-1} and e are chosen the same size:

$$r^{-1} > \sqrt{\frac{p}{2}} \quad \wedge \quad e > \sqrt{\frac{p}{2}}$$

However, this condition would require the key to become too small for the protocol to be secure:

$$\begin{aligned} e < a \quad \wedge \quad a \times k < b \quad \wedge \quad b \times r^{-1} < p \\ e \times k < b \quad \wedge \quad b < \frac{p}{r^{-1}} \\ e \times k < \frac{p}{r^{-1}} \\ k \times e \times r^{-1} < p \\ k \times \frac{p}{2} < p \\ k < 2 \end{aligned}$$

With only the possibility 0 and 1 for the key, the key can be easily recovered from c_1 .

5 Conclusion

We presented two efficient MitM attacks and four key recovery attacks against QP-KEM. We proposed countermeasures against all but the lattice reduction attack. The MitM attacks can be prevented by signing and verifying the first two exchanged values. The brute force key recovery attacks, the most effective one having a computational complexity of 2^{40} , can be prevented by increasing parameters in the protocol. However, we see no way to prevent the key recovery attack that uses lattice reduction.

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